UNIT 2

Exercise 1: Zero-shot, One-shot, and Few-shot Prompting for a Smart Fitness Watch

Aim

To design and compare zero-shot, one-shot, and few-shot prompts for generating a product description for a Smart Fitness Watch.

Algorithm

- Define the product and its features.
- Create a zero-shot prompt without examples.
- Create a one-shot prompt with one example.
- Create a few-shot prompt with multiple examples.
- Display and compare the generated descriptions.

Python Code

```
def generate_description(prompt_type):
    if prompt_type == "zero-shot":
        prompt = """
        Write a product description for a Smart Fitness Watch.
        Include its features and benefits.
        """
    elif prompt_type == "one-shot":
        prompt = """
        Example:
        Product: Wireless Earbuds
        Description: Enjoy clear sound and comfortable listening.

        Product: Smart Fitness Watch
        Write a similar product description mentioning its features.
        """
    else:
        prompt = """
        Example 1:
        Product: Smart Bottle
        Description: Stay hydrated with convenient water reminders.

        Example 2:
        Product: Fitness Band
        Description: Track your daily steps and calories easily.

        Product: Smart Fitness Watch
        Write a similar product description.
        """
```

Sample Output

```
Prompt Type: zero-shot
Prompt:
        Write a product description for a Smart Fitness Watch.
        Include its features and benefits.


Prompt Type: one-shot
Prompt:
        Example:
        Product: Wireless Earbuds
        Description: Enjoy clear sound and comfortable listening.

        Product: Smart Fitness Watch
        Write a similar product description mentioning its features.


Prompt Type: few-shot
Prompt:
        Example 1:
        Product: Smart Bottle
        Description: Stay hydrated with convenient water reminders.

        Example 2:
        Product: Fitness Band
        Description: Track your daily steps and calories easily.

        Product: Smart Fitness Watch
        Write a similar product description.
```

Result

The three prompt types were designed successfully. Zero-shot uses no examples, one-shot uses one example, and few-shot uses multiple examples to guide the product description.